


```

import cv2 import
numpy as np def
main():

    video_path = "t1.mp4"    # keep video in same folder

cap = cv2.VideoCapture(video_path)

    if not cap.isOpened():
print("Error: Cannot open video")
return

    ret, first_frame = cap.read()    if
not ret:        print("Error: Cannot
read video")        return

    prev_gray = cv2.cvtColor(first_frame, cv2.COLOR_BGR2GRAY)
prev_gray = cv2.GaussianBlur(prev_gray, (15,15), 0)
print("Press 'q' to quit")

    while True:        ret,
frame = cap.read()        if
not ret:        break

        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
gray = cv2.GaussianBlur(gray, (15,15), 0)        diff
= cv2.absdiff(prev_gray, gray)

        thresh = cv2.threshold(diff, 15, 255, cv2.THRESH_BINARY)[1]
thresh = cv2.dilate(thresh, None, iterations=3)

        contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)

        for c in contours:
if cv2.contourArea(c) > 200:
            x,y,w,h = cv2.boundingRect(c)
            cv2.rectangle(frame, (x,y), (x+w,y+h), (0,255,0), 2)

cv2.imshow("Motion", frame)        prev_gray = gray

        if cv2.waitKey(30) & 0xFF == ord('q'):
break

```

```
cap.release()  
cv2.destroyAllWindows()  
  
if __name__ == "__main__":  
    main()  
    Press 'q' to quit
```